

Threat Intelligence Report: Amadey Persistent Botnet Activity

- **Report ID:** 2026-05-19-AMA-01
- **Classification:** TLP:CLEAR
- **Subject:** Manual Cypher Deobfuscation and Dynamic C2 Extraction of Amadey Botnet

Executive Summary

Detailed static and dynamic analysis was conducted on a native 64-bit Windows executable compiled to run the Amadey botnet payload. The sample uses an aggressive installation routine to copy itself into system temporary folders under randomised names while utilising a watchdog loop to establish persistence. The malware successfully evades automated string deobfuscation by processing its primary indicators through a custom alphanumeric substitution cypher calculated dynamically in memory. Dynamic attachment and API hooking successfully extracted live network indicators.

List of Affected Entities

Indicator Field	Operational Entity Data
Internal Source (Assigned)	Analyst-VM-01 (Host Operating System / User Account: Analyst)
Target Identity	Amadey Command and Control Server
Attacker IP	196.251.107.130
Identity	Unknown External Actor
VPN Gateway	None (Direct HTTP outbound transport detected)

Table 1: List of Affected Entities

Technical Analysis & Host Indicators

1. Dropper & Persistence Mechanism

Upon initial execution, the malware conducts a path validation check via `GetModuleFileNameA`. If the file is executed from an external directory (e.g., a Desktop environment), it deploys its persistent instance into the user's local profile before executing the clone and terminating the dropper instance.

- **Persistent Host File Path:**
`C:\Users\Analyst\AppData\Local\Temp\08e25196df\Ujkjwrro.exe`
- **Process Behaviour:** The persistent application runs an automated watchdog loop. If the task is killed via administrative termination, the operating system persistence keys instantly trigger a restart of the `Ujkjwrro.exe` process.

2. Cryptographic Architecture

The configuration data blocks are obfuscated utilising a custom Vigenère-style shift cypher. The alphabet array used as the base string for the character transformation is uninitialized within the static file sections (Base Address: `14007c858`). It is dynamically generated in memory at runtime via module `FUN_14005f180`. Decryption is processed through a modular arithmetic loop structured as:

Decrypted Character = (Ciphertext Index - Key Index + Alphabet Length) (mod Alphabet Length)

3. Network Indicators & Infrastructure Binding

Network communications are established via **WININET.dll** functions. The malware uses hardcoded callback intervals to beacon operational data back to the centralised attacker node. Dynamic register tracking at the **InternetConnectA** API entry point successfully exposed the following indicators of compromise:

- **Target Destination IP Address:** 196.251.107.130 (Mapped to active external C2 host)
- **HTTP Campaign Callback Path:** /h84jjfAr/index.php
- **Botnet Campaign Tracker ID:** 3198791665
- **User-Agent String Signature:** Configured dynamically via **HttpOpenRequestA** parameters.
- **SHA256:** 5258a241f8c67e1666060e2c033d127dee8c2c50c7a8dfe0676264e9cb6762c1

Technical Resources

- **Detection Engineering:** [View the custom YARA rule here](#)
- **IoC Publication:** [Access the full list of IoCs on AlienVault OTX](#)